Program

```
#include <stdio.h>
#define MAX 50
int main()
{
   int graph[MAX][MAX];
   int visited[MAX];
   int queue[MAX];
   int n, i, j;
   int front = 0, rear = 0;
   int start, current;
   // Read number of vertices
   printf("Enter number of vertices: ");
   scanf("%d", &n);
   // Read adjacency matrix
   printf("Enter the adjacency matrix:\n");
   for (i = 0; i < n; i++)
   {
      for (j = 0; j < n; j++)
      {
         scanf("%d", &graph[i][j]);
      }
   }
   // Initialize visited array
   for (i = 0; i < n; i++)
   {
     visited[i] = 0;
   }
   // Read starting vertex
   printf("Enter starting vertex: ");
   scanf("%d", &start);
   // Insert starting vertex into queue
   visited[start] = 1;
   queue[rear] = start;
   rear++;
   printf("BFS Traversal: ");
   // BFS traversal
   while (front < rear)
  {
      // Remove vertex from queue
      current = queue[front];
      front++;
      printf("%d ", current);
```

```
    // Find adjacent vertices
    for (i = 0; i < n; i++)
    {
      if (graph[current][i] == 1 && visited[i] == 0)
      {
        visited[i] = 1;
        queue[rear] = i;
        rear++;
       }
    }
  }
  return 0;
}
```

Sample input

Enter number of vertices: 5
Enter the adjacency matrix:
0 1 1 1 0
1 0 1 0 0
1 1 0 0 1
1 0 0 0 0
0 0 1 0 0
Enter starting vertex: 0

Output

BFS Traversal: 0 1 2 3 4